



UHTS-DRL: A deep reinforcement learning framework for integrated agile satellite observation and data transmission scheduling

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ABSTRACT

Efficient Earth observation satellite scheduling requires integrated management of both observation and data transmission tasks to optimize the overall system efficiency in practice. However, conventional approaches typically treat these processes separately, leading to suboptimal resource utilization. This paper formulates the Earth observation satellite scheduling as a Multi-agile Satellite and Multi-ground Station Integrated Observation and Data Transmission Scheduling Problem (MSIODTSP) and proposes a Unified Hierarchical Two-stage Scheduling Deep Reinforcement Learning (UHTS-DRL) framework for solving MSIODTSP. The UHTS-DRL approach formulates MSIODTSP as a unified Markov Decision Process (MDP), enabling joint optimization of observation and transmission scheduling while accounting for multi-orbit time windows and complex operational constraints. The framework employs a hierarchical policy network comprising a time window encoder, a state encoder, and a dual-decoder for two-stage scheduling, facilitating end-to-end optimization without the need for handcrafted heuristics. Experimental results demonstrate that UHTS-DRL consistently outperforms existing approaches across various problem scales, resource configurations, and geographical target distributions, achieving up to 12.5% relative improvement in total profit while maintaining computational efficiency.

1. Introduction

The increasing demand for Earth observation services, driven by applications in environmental monitoring, disaster response, and military reconnaissance, has led to significant challenges in data acquisition [1]. To address these challenges, agile satellites have been widely deployed in Earth observation networks. Unlike conventional satellites limited to nadir observations, agile satellites can rotate along multiple axes to observe targets at various angles, transforming discrete observation points into extended observation time windows (OTWs) [2]. However, this agility introduces time-dependent transition constraints between consecutive observations, as the required maneuvering time varies based on the angular distance between targets, significantly complicating the scheduling problem.

While agile satellites enhance data collection capabilities, they must also periodically transmit the acquired data to Earth via ground station networks during specific transmission time windows (TTWs). The scheduling complexity increases substantially when

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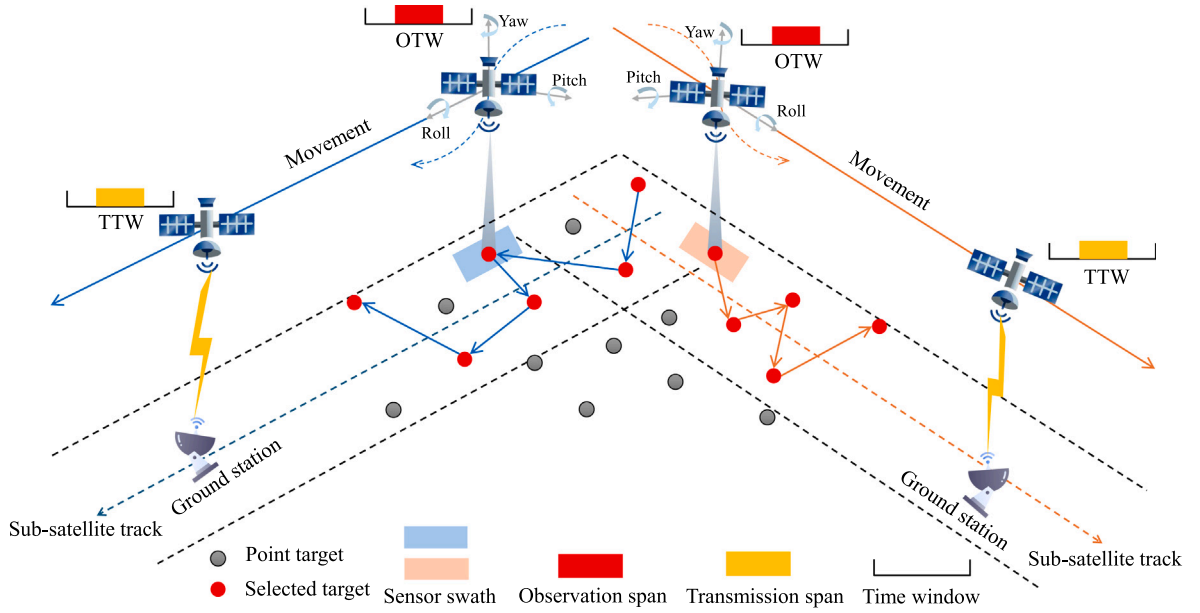


Fig. 1. Schematic diagram of MSIODTSP. There are two satellites observing their respective targets. Red dots represent selected targets for observation, while gray dots indicate unobserved targets.

considering that satellites could perform multiple orbits around the Earth within a time horizon, generating numerous OTWs and TTWs [2,3]. This creates an intricate operational environment where observation and data transmission tasks must be coordinated while simultaneously satisfying multiple constraints, including visibility windows, onboard storage limitations and time-dependent transition constraints. In addition, joint observation-transmission scheduling enables optimal resource allocation across the entire scheduling horizon, maximizing system throughput and observation benefits while minimizing resource conflicts and idle periods. Consequently, developing efficient algorithms for integrated observation and transmission scheduling represents a critical research direction for enhancing satellite mission productivity.

A critical limitation of existing approaches lies in their fragmented treatment of observation and data transmission tasks. Most prior studies focus exclusively on either satellite observation scheduling [4,5] or data transmission scheduling [6], neglecting the inherent interdependencies between these two processes in real-world applications. This decoupled optimization leads to suboptimal resource utilization: aggressively scheduling observation tasks without coordinated data downlink planning may result in onboard memory saturation, while scheduling too many ground station contacts could waste potential observation opportunities due to overlapping time windows [7]. Therefore, to maximize the overall Earth observation system efficiency while maintaining realism in the formulation of the scheduling problem, we study the Multi-agile Satellite and Multi-ground Station Integrated Observation and Data Transmission Scheduling Problem (MSIODTSP).

As shown in Fig. 1, MSIODTSP involves scheduling multiple agile satellites, which can adjust their orientation to observe different targets, and multiple ground stations that receive data from satellites. Each satellite must complete observations within its OTW, and subsequently transmit the collected data to ground stations during the TTW. The objective is to maximize the total profit from completed tasks, where a task is considered fully completed only when both relevant observation and data transmission are successfully executed. Partial profit is assigned to tasks that are observed but not transmitted, reflecting the value of partial completion.

In the past few years, various approaches have been reported including exact methods [2,8], heuristic methods [9,10] and meta-heuristic methods [7,11] for solving MSIODTSP. However, there is a critical trade-off between solution quality and computational efficiency, where exact methods become intractable for large instances while heuristics and metaheuristics sacrifice optimality for speed. Deep reinforcement learning (DRL) presents a promising alternative for solving combinatorial optimization problems by learning effective decision policies from patterns in extensive instance data with high computational efficiency [12,13]. While DRL approaches have been applied to satellite observation scheduling [3], applications specifically addressing integrated observation and data transmission scheduling remain limited. Existing DRL-based approaches for integrated observation and data transmission scheduling incorporate several simplifications that limit their practical applicability, such as considering only single satellites in single orbits [14] and simplifying satellites' resource limitations [15,16] which do not fully reflect the multifaceted nature of real-world satellite operations.

To address the limitations of previous studies by considering multi-agile satellite, multi-ground station scenarios while fully accounting for various constraints, we propose UHTS-DRL (Unified Hierarchical Two-stage Scheduling Deep Reinforcement Learning), an end-to-end framework for solving MSIODTSP.

We first formulate a unified Markov Decision Process (MDP) model for MSIODTSP, and then design a hierarchical policy network, which features three key components: a time window encoder, a state encoder, and a dual decoder. The main contributions of this paper are summarized as follows:

- We introduce the MSIODTSP, which focuses on jointly optimizing observation and data transmission operations for Earth observation satellites. To address this challenge, we establish a unified MDP model that effectively captures the sequential decision-making nature of satellite operations while considering critical constraints such as time windows, resource limitations, and operational dependencies.
- We introduce a UHTS-DRL framework that employs a hierarchical neural network, composed of a time window encoder, a state encoder, and a dual-decoder, to solve the MSIODTSP in an end-to-end manner. Specifically, the time window encoder, based on a Long Short-Term Memory (LSTM) network, effectively encodes the multiple time windows generated by multi-orbit satellite operations; the state encoder captures the dynamic elements of the decision-making process; and the dual-decoder, enhanced with attention mechanisms, reduces the problem's complexity by decomposing decisions into sequential satellite and task selections. This architecture addresses the MSIODTSP without relying on handcrafted heuristics or expert knowledge. To the best of our knowledge, this is the first work to tackle the MSIODTSP using a DRL-based approach.
- Extensive experiments across various scenarios demonstrate the superior performance of UHTS-DRL compared to traditional methods. The results indicate that UHTS-DRL consistently achieves higher profit rates while maintaining computational efficiency across different problem scales, resource configurations and geographical target distributions.

This paper is organized as follows: [Section 2](#) reviews related work on DRL-based satellite scheduling. [Section 3](#) introduces the MSIODTSP problem. [Section 4](#) details the unified MDP model for MSIODTSP, while [Section 5](#) describes the proposed solution. [Section 6](#) presents the experimental results, and [Section 7](#) concludes the paper and outlines future directions.

2. Related work

This section focuses on the application of RL in satellite scheduling problems. The content is organized into three main categories: satellite observation scheduling, satellite data transmission scheduling, and integrated satellite observation and data transmission scheduling.

2.1. Satellite observation task scheduling

Research in satellite observation scheduling has progressed significantly at both single-satellite and multi-satellite levels in recent years. For single-satellite scheduling, the emergence of deep reinforcement learning (DRL) has enabled more adaptive and intelligent scheduling solutions. Generic MDP formulations [17] have demonstrated strong adaptability to unseen scenarios. To better handle complex spatial relationships, local attention mechanisms [18] have been integrated into DRL frameworks to capture dependencies between observation targets. To address the challenge of enforcing hard constraints in purely learning-based methods, hybrid approaches integrating DRL with rule-based methods [19] have been developed. Such approaches have demonstrated practical value in operational satellite systems, exemplified by the FIS-PPO algorithm [20] which achieved improved observation efficiency for the SATech-01 satellite while ensuring fair resource allocation.

In multi-satellite scheduling, coordination among multiple platforms substantially increases complexity, necessitating advanced neural architectures. Graph neural networks combined with Transformer architectures [21] effectively capture inter-satellite relationships and long-range dependencies, while multi-pointer networks with attention mechanisms [22] provide scalability for variable constellation sizes. Single-agent reinforcement learning approaches [23] have shown superior performance in free communication scenarios with simpler training procedures. To address heterogeneous satellite systems, specialized solutions combining DRL with genetic algorithms [24] and multi-objective learning frameworks [25] have been developed. For missions with specific temporal patterns, dedicated algorithms such as POMP [14] improve efficiency by exploiting task periodicity. Large-scale constellations motivate hierarchical approaches [26] that separate daily allocation from emergency response. Additionally, DRL-based frameworks [27] have been proposed for coordinating satellites across both high and low Earth orbits.

2.2. Satellite data transmission scheduling

In satellite data transmission scheduling, Ren et al. [28] developed a specialized framework for high-altitude satellites and Ou et al. [29] combined DRL with heuristic approaches for large-scale scenarios. In satellite data transmission networks with different architectures such as data relay satellite networks and low Earth orbit satellite networks, researchers have proposed various DRL-based scheduling solutions. For data relay satellite networks, Li et al. [30] developed an attention-based encoder-decoder policy network showing excellent scalability. Ma et al. [31] tackled telemetry, tracking, and command (TT&C) mission scheduling using DRL, while Li et al. [32] developed a twin delayed deep deterministic policy gradient-based approach for the online satellite data transmission scheduling problem. Chen et al. [33] introduced the RL-based ground station scheduling algorithm specifically for low Earth orbit satellites, addressing both batch and individual task planning.

2.3. Integrated observation and transmission scheduling

The integration of observation and transmission scheduling remains a relatively unexplored area in satellite task scheduling. Early attempts to address this challenge include Qu et al. [34] who developed an intelligent scheduling framework using imitation learning based on mixed integer linear programming. For single satellite scenarios, Wen et al. [35] proposed a hybrid actor-critic reinforcement learning method to handle intensive observation scenarios. Long et al. [36] introduced a deep reinforcement learning approach with varying-scale generalization for Earth observation satellite scheduling that comprehensively addresses resource consumption and supplementary constraints. In terms of multi-satellite approaches, Chen et al. [15] applied multi-agent DRL with decentralized partially observable MDP framework, while Li et al. [16] developed an attention-based distributed planning algorithm to coordinate mission execution across multiple satellites, but ignored ground station availability constraints where multiple satellites might compete for limited downlink opportunities. Although these approaches offer valuable contributions, they have either focused only on non-agile satellites or simplified the agility of agile satellites in their modeling frameworks.

Some critical limitations in satellite scheduling have been identified through the above literature review. First, most approaches separate observation and transmission scheduling into distinct optimization phases. Second, in the integrated observation and transmission scheduling research, most studies only consider non-agile satellites and incorporate simplifications in their planning, such as neglecting time-dependent transition times and ground station resource constraints, which cannot effectively reflect real-world operational conditions. These limitations highlight the need for more comprehensive approaches that effectively model the coupled problem while maintaining computational efficiency.

3. Problem description

The MSIOTSP involves coordinating observation and data transmission activities across multiple agile satellites and ground stations within a specified planning horizon. In this problem, a set of satellites S needs to observe a set of targets T and transmit the collected data to multiple ground stations G within the planning horizon H .

As mentioned above, agile satellites possess the capability to perform attitude maneuvers that enable them to observe targets at varying angles off their ground track. This agility significantly expands the visible time windows for target observation, but also introduces additional complexity to the scheduling problem, as the satellite requires time to transition between different observation angles. Each agile satellite can rotate around three axes (roll, pitch, and yaw) to adjust its attitude for observing different targets, with the transition time between targets dependent on the angular distance between their observation positions.

Each target may have multiple visible time windows for both observation and data transmission during the planning horizon, with the maximum number of windows limited to N_{max} . A target could be observed by an agile satellite during its visible observation windows, and its data could be transmitted to one ground station during the visible transmission windows. For each target, its observation must be completed before its corresponding data transmission can begin. Each satellite has limited onboard resources: a maximum memory capacity M_{max} that constrains the total storage of observed tasks' data, and a maximum energy capacity E_{max} for each orbital period T_{orbit} . The energy of the satellites is reset to E_{max} at the beginning of each new orbital period.

While a target's mission receives a full profit only when both observation and subsequent data transmission are achieved, targets that are observed but not yet transmitted are still included in the objective function with a partial profit. Accounting for these observation-only targets allows for better use of available observation time windows, which might otherwise be underused if only fully completed missions were considered. This partial profit setting (similar to the partial objective function value in [37]) is also reasonable in practice, since the observation data stored on the satellite could potentially be transmitted to a ground station after the planning horizon.

4. Unified MDP model

As shown in Fig. 2, the UHTS-DRL is divided into two main components: the unified MDP model and the policy network. The unified MDP model is defined by a 4-tuple $\mathcal{M} = \{S, \mathcal{A}, \mathcal{P}, \mathcal{R}\}$. The state space S is comprehensively represented by static features and dynamic features. The action space $\mathcal{A}$ consists of a selected satellite i and a selected task j . Actions generated by the policy network influence the state through transition rules $\mathcal{P}$, which in turn yield corresponding rewards through reward function $\mathcal{R}$. In terms of the policy network, the component consists of state and time window encoders that process the problem instance, and a dual-decoder containing satellite selector and task selector modules that produce the actions. The reward signals obtained from the unified MDP model are utilized by the REINFORCE algorithm to train the policy network, creating a reinforcement learning loop that optimizes satellite task scheduling decisions.

4.1. State

The state space S is designed to capture all essential information required for satellite scheduling decisions, consisting of static features that remain unchanged throughout the scheduling process and dynamic features that evolve with scheduling decisions.

4.1.1. Static features

The static features comprise three essential components: observation windows $\mathbf{W}_{obs}$, transmission windows $\mathbf{W}_{trans}$, and attitude requirements Θ . Specifically, each observation window $w_{obs}^{ijl} \in \mathbf{W}_{obs}$ defines a feasible time interval $[t_{start}^{ijl}, t_{end}^{ijl}]$ during which the satellite i can observe the target j in its l -th window. Similarly, each transmission window $w_{trans}^{ikl} \in \mathbf{W}_{trans}$ specifies an interval

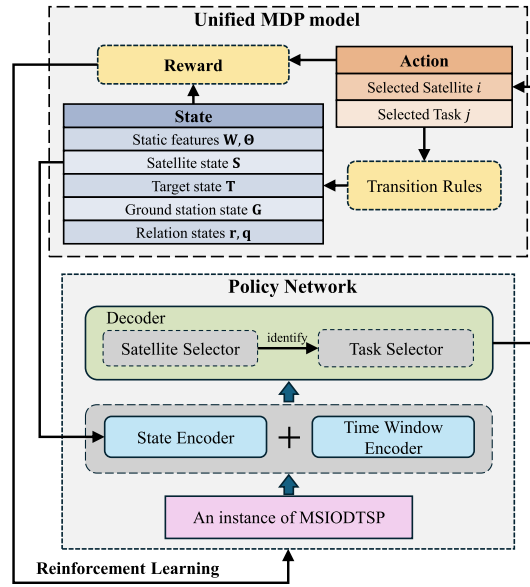


Fig. 2. The framework of solving MSIODTSP using UHTS-DRL.

Table 1
Satellite state.

Attributes	Description	Range
e_i	Remaining energy capacity	$[0, E_{max}]$
m_i	Available memory storage	$[0, M_{max}]$
t_i	Last task completion time (observation or transmission)	$[0, H]$
θ_i	Current Euler angles $[\theta_i, \phi_i, \psi_i]$	$\mathbb{R}^3$
n_i	Number of stored tasks in memory	$\{0, 1, \dots, T \}$
p_i	Remaining time of current orbital period	$[0, T_{orbit}]$
α_i	Number of total remaining observation windows across all tasks	$\mathbb{N}_{>0}$
β_i	Number of total remaining transmission windows across all tasks	$\mathbb{N}_{>0}$
v_i	Total value of waiting tasks	$\mathbb{R}_{\geq 0}$
ω_i	Average remaining transmission windows	$\mathbb{R}_{\geq 0}$
τ_i^o	Time to the nearest observation window	$[0, H]$
τ_i^t	Time to the nearest transmission window	$[0, H]$

$[t_{start}^{ikl}, t_{end}^{ikl}]$ when the satellite i can transmit data to the ground station k in its l -th window. For each observation window, the attitude requirements $\theta_{ijl}(t) \in \Theta$ prescribe a series of Euler angles (roll, pitch, and yaw) $[\theta(t), \varphi(t), \psi(t)]$ that satellite i must maintain at each time point $t \in [t_{start}^{ijl}, t_{end}^{ijl}]$ while observing target j , ensuring precise attitude control throughout the duration of observation. These static features collectively define the fundamental temporal, spatial, and operational constraints that must be met throughout the scheduling process.

4.1.2. Dynamic features

The dynamic state of the scheduling system at decision time t is characterized by five key components: satellite states S_i^t , target states T_j^t , ground station states G_k^t , satellite–target relations r_{ij}^t , and satellite–ground station relations q_{ik}^t . These states are continuously updated through a state transition function to reflect the evolution of the system during the scheduling process, enabling real-time monitoring and adaptive management of operations. For notational simplicity in the following discussion, we omit the time index t when describing the states and their transitions, with the understanding that all state variables are time-dependent.

Satellite State: Each satellite state, represented by a 14-dimensional vector $S_i \in \mathbb{R}^{14}$, encapsulates comprehensive information about the operational status of satellite i . This includes resource levels, temporal status, attitude configuration, and relevant task metrics. The details of the satellite state are listed in Table 1.

Target State: The target state $T_j \in \mathbb{R}^7$ maintains the essential properties and the execution status of each target j . This state tracks both the intrinsic requirements of the target and its progress through the observation and transmission phases. The details of the target state are listed in Table 2.

Ground Station State: Ground station states $G_k \in \mathbb{R}^2$ record the operational history of each ground station, maintaining information about their most recent data transmission activities. The details of the ground station state are listed in Table 3.

Table 2
Task state.

Attributes	Description	Range
π_j	Target profit	$\mathbb{R}_{>0}$
δ_j	Required observation duration	$\mathbb{R}_{>0}$
μ_j	Storage requirement	$\mathbb{R}_{>0}$
o_j	Observation completion indicator (0:no, 1:yes)	$\{0, 1\}$
σ_j	Observing satellite	$\{0, \dots, S \}$
e_j	Observation completion time	$[0, H]$
c_j	Transmission completion indicator (0:no, 1:yes)	$\{0, 1\}$

Table 3
Ground station state.

Attributes	Description	Range
λ_k	Last transmission task completion time	$[0, H]$
γ_k	Last communicating satellite	$\{0, \dots, S \}$

Table 4
Satellite–target relationship.

Attributes	Description	Range
η_{ij}	Earliest possible observation time	$[0, H]$
v_{ij}	Number of remaining observation windows	$\{0, 1, \dots, N_{max}\}$

Table 5
Satellite–station relationship.

Attributes	Description	Range
ζ_{ik}	Earliest possible transmission time	$[0, H]$
κ_{ik}	Number of remaining transmission windows	$\{0, 1, \dots, N_{max}\}$

In addition to the states mentioned above, we also maintain two types of relational states that capture the temporal and resource constraints between components. These states are crucial for scheduling feasible operations.

Satellite–Target Relation: Relations $r_{ij} \in \mathbb{R}^2$ between satellites and targets track the feasibility and availability of observation opportunities. The details of the satellite–target relation are listed in Table 4.

Satellite–Station Relation: Relations $q_{ik} \in \mathbb{R}^2$ between satellites and ground stations monitor the availability of data transmission opportunities. The details of the satellite–station relation are listed in Table 5.

4.2. Action

The action space $\mathcal{A}$ comprises the selection of satellites and their operational tasks at each decision step. Specifically, an action $a_t \in \mathcal{A}$ is represented as a tuple (i, j) , where i indicates the selected satellite and j specifies the task. The task selection encompasses three types of operations. First, when an observation target is selected, the satellite performs observation operations subject to visibility windows, energy capacity and memory constraints. Second, if a ground station is selected, the satellite transmits its stored observed tasks during a feasible communication window. Lastly, to handle situations where satellites have insufficient energy for any operational tasks, we introduce a virtual task option that advances the satellite to its next orbital period, resetting its energy to maximum capacity and updating other parameters. The execution time for each action is determined by the earliest feasible start time, which is stored in satellite–target relations r or satellite–station relations q . The earliest start times are dynamically recalculated when satellite or ground station states change, taking into account both satellite maneuverability constraints and time window availabilities.

4.3. Transition function

The transition function $\mathcal{P}$ defines the evolution of system states after executing an action. Given current state s_t and action $a_t = (i, j)$, the state updates follow different patterns according to the operation type.

For observation operations, j in the action represents an observation target. The satellite's operational parameters are updated to reflect both resource consumption and task execution. The primary state transitions include:

$$e_i \leftarrow \begin{cases} E_{max} - (t_i + \delta_j - T_{next}) \cdot E_{rate}, & \text{if cross orbit} \\ e_i - \delta_j \cdot E_{rate}, & \text{otherwise} \end{cases} \quad (1)$$

$$m_i \leftarrow m_i - \mu_j \quad (2)$$

$$t_i \leftarrow \eta_{ij} + \delta_j \quad (3)$$

$$n_i \leftarrow n_i + 1 \quad (4)$$

$$p_i \leftarrow T_{orbit} - ((\eta_{ij} + \delta_j) \bmod T_{orbit}) \quad (5)$$

where T_{next} represents the start time of next orbital period, E_{rate} is the energy consumption rate. The satellite's attitude angles θ_i are adjusted to match the required observation geometry. The system then recalculates window availability metrics (α_i , β_i) and target-related parameters (v_i , ω_i , τ_i^o , τ_i^f) based on the updated state. Correspondingly, target j updates its status vector as:

$$\begin{cases} o_j \leftarrow 1 \\ \sigma_j \leftarrow i \\ \epsilon_j \leftarrow \eta_{ij} + \delta_j \end{cases} \quad (6)$$

Following the satellite state update, the relational states for satellite i must also be updated. The earliest execution times in r_{ij} and q_{ik} are recalculated considering visibility windows, attitude maneuver times, and resource availability. This ensures the time-dependent constraints of agile satellites are satisfied. Meanwhile, ground station states remain unchanged during observation operations.

For data transmission operations, j in the action represents a ground station. Satellite's state transitions primarily reflect data transfer and resource consumption:

$$e_i \leftarrow \begin{cases} E_{max} - (t_i + \frac{\delta_j}{\rho} - T_{next}) \cdot E_{rate}, & \text{if cross orbit} \\ e_i - \frac{\delta_j}{\rho} \cdot E_{rate}, & \text{otherwise} \end{cases} \quad (7)$$

$$m_i \leftarrow m_i + \mu_j \quad (8)$$

$$t_i \leftarrow \zeta_{ij} + \frac{\delta_j}{\rho} \quad (9)$$

$$n_i \leftarrow n_i - 1 \quad (10)$$

$$p_i \leftarrow T_{orbit} - ((\zeta_{ij} + \frac{\delta_j}{\rho}) \bmod T_{orbit}) \quad (11)$$

where ρ denotes the transmission-to-observation time ratio. Given the scarcity of transmission windows during the planning horizon, the system prioritizes task transmission based on their profits. Specifically, targets stored in satellite i are sorted in descending order of their profits and transmitted sequentially until either all targets are transmitted or constraints (transmission window duration, energy) are reached. For each transmitted target m that was previously observed by satellite i , its status updates as:

$$c_m \leftarrow 1 \quad (12)$$

The ground station j records the completion of transmission by updating:

$$\begin{cases} \lambda_j \leftarrow \zeta_{ij} + \frac{\delta_j}{\rho} \\ \gamma_j \leftarrow i \end{cases} \quad (13)$$

Following the satellite state update, the relational states r_{im} and q_{ij} for satellite i are also recalculated considering the data transmission preparation time constraints between satellites and ground stations.

As for virtual task operations, designed to handle energy depletion scenarios, the satellite advances to its next orbital period while resetting its operational parameters:

$$e_i \leftarrow E_{max} \quad (14)$$

$$t_i \leftarrow \min[(\lfloor t_i / T_{orbit} \rfloor + 1) \cdot T_{orbit}, H] \quad (15)$$

$$p_i \leftarrow \begin{cases} 0 & \text{if } t_i = H \\ T_{orbit} & \text{otherwise} \end{cases} \quad (16)$$

Target states and ground station states remain unchanged, while the relational states are recalculated due to the new state of the satellite.

It is important to note that while our current implementation uses deterministic parameters for benchmark comparison, the transition function $\mathcal{P}$ naturally accommodates stochastic execution times by updating states based on realized values at each decision step. This design allows UHTS-DRL to extend to uncertain operational environments without architectural changes.

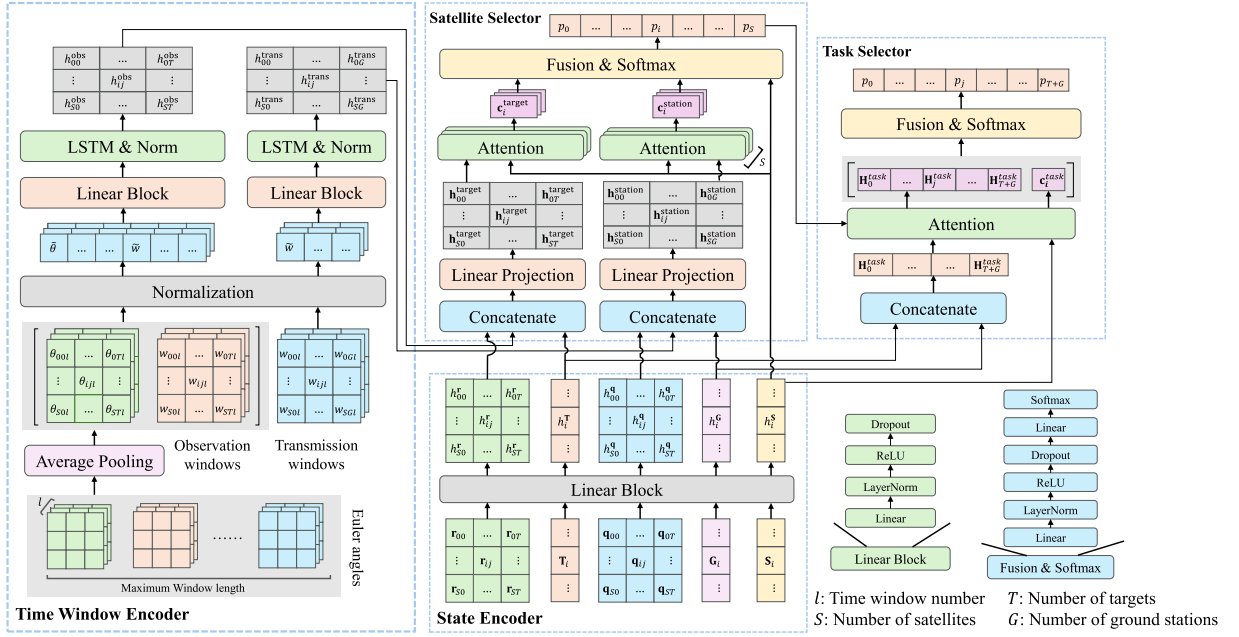


Fig. 3. The architecture of the policy network.

4.4. Reward

The reward function $\mathcal{R} : S \times \mathcal{A} \rightarrow \mathbb{R}$ is defined as:

$$\mathcal{R}(s, a) = \begin{cases} 0 & \text{for virtual task} \\ w_j^o \cdot \pi_j & \text{for observation completion} \\ \pi_j & \text{for transmission completion} \end{cases} \quad (17)$$

where w_j^o represents the lower weight profit for target j when only completing observation.

5. Methodology

In this section, we introduce the proposed policy network for solving MASIODTSP. Our architecture incorporates attention mechanisms and LSTM-based sequential processing to handle the complex relationships among satellites, tasks, and ground stations. The architecture of the policy network is illustrated in Fig. 3.

5.1. Overview

The policy network models the schedule construction as a sequential decision process where at each step, both a satellite and a task (a target for observation or a ground station for transmission) are selected. Specifically, we focus on learning a stochastic policy $\pi_\theta(a_t | s_t)$ represented by a neural network with trainable parameters θ . The joint probability of generating a complete schedule can be factorized based on the chain rule:

$$p(s_T | s_0) = \prod_{t=0}^{T-1} \pi_\theta(a_t | s_t) p(s_{t+1} | s_t, a_t) \quad (18)$$

where s_0 denotes the initial schedule state, and s_T represents the terminal state.

The policy network architecture consists of three main components: a time window encoder for processing observation and transmission windows, a state encoder for capturing system dynamic states, and a two-stage scheduling decoder that enables both satellite selection and target selection at each decision step. Specifically, the time window encoder processes the observation and transmission windows once at the beginning, since these time windows remain unchanged throughout the scheduling process. The state encoder, however, needs to re-encode the state information after each decision as the states of satellites, tasks, and stations are dynamically updated. At each decision step t , the decoder first selects a satellite i based on the current encoded states and window features, then selects a feasible task j (either a target or ground station) for the chosen satellite, forming the action $a_t = (i, j)$. This process continues until either all tasks are scheduled or all satellites reach the planning horizon H . In the following subsections, we elaborate on each component of this architecture.

5.2. Time window encoder

The time window encoder embeds observation windows with angle constraints and transmission windows into a high-dimensional feature space. Given the different characteristics of these windows, we design specialized encoding mechanisms for each type.

To enhance model generalization, we first normalize the time windows:

$$\tilde{w}_{ijl}^{start} = \frac{w_{ijl}^{start}}{H} \quad (19)$$

$$\tilde{w}_{ijl}^{end} = \frac{w_{ijl}^{end}}{H} \quad (20)$$

where $\tilde{w}_{ijl}$ represents a normalized time window of the l -th window for satellite i observing or transmitting task j , and H is the planning horizon.

For observation windows, considering that each satellite–task pair may have multiple available time windows with varying attitude requirements, we design an LSTM-based encoding structure. First, the normalized time windows and Euler angles are encoded through separate neural networks, each consisting of a linear projection, layer normalization, ReLU activation, and dropout layers:

$$e_{ijl}^{time} = \text{Dropout}(\text{ReLU}(\text{LayerNorm}(\text{FF}(\tilde{w}_{ijl})))) \quad (21)$$

$$e_{ijl}^{angle} = \text{Dropout}(\text{ReLU}(\text{LayerNorm}(\text{FF}(\tilde{\theta}_{ijl})))) \quad (22)$$

Here, $\tilde{\theta}_{ijl}$ represents the average Euler angles (roll, pitch, and yaw) that define the satellite's required attitude during observation window w_{ijl} .

To process the sequence of available windows and their associated attitude requirements, we employ an LSTM network that takes the concatenated features as input:

$$h_{ij}^{obs} = \text{LSTM}\left(\left[e_{ijl}^{time}, e_{ijl}^{angle}\right]\right) \quad (23)$$

where the final hidden state h_{ij}^{obs} represents the encoded observation window features for this satellite–target pair.

For transmission windows, which only require temporal accessibility constraints without attitude constraints, we implement a simpler encoding process:

$$h_{ij}^{trans} = \text{LSTM}(e_{ijl}^{time}) \quad (24)$$

the final hidden state h_{ij}^{trans} is taken as the transmission window feature of satellite i to ground station j .

5.3. State encoder

The state encoder is designed to capture the dynamic features of the scheduling system. We encode all the dynamic states: satellite states S_i , target states T_i , station states G_i , and relation states r_{ij} (satellite–target) and q_{ij} (satellite–station). Each type of state is processed through an MLP encoder:

$$\begin{cases} h_i^S = \text{Dropout}(\text{ReLU}(\text{LayerNorm}(\text{FF}(S_i)))) \\ h_i^T = \text{Dropout}(\text{ReLU}(\text{LayerNorm}(\text{FF}(T_i)))) \\ h_i^G = \text{Dropout}(\text{ReLU}(\text{LayerNorm}(\text{FF}(G_i)))) \\ h_{ij}^r = \text{Dropout}(\text{ReLU}(\text{LayerNorm}(\text{FF}(r_{ij})))) \\ h_{ij}^q = \text{Dropout}(\text{ReLU}(\text{LayerNorm}(\text{FF}(q_{ij})))) \end{cases} \quad (25)$$

This encoding process is performed through two consecutive layers, yielding the final encoded representations that capture the dynamic characteristics of the system states.

5.4. Dual-decoder

The dual-decoder implements a hierarchical decision-making process that sequentially selects both satellites and the corresponding tasks (observation targets or ground stations) at each decision step. It consists of two main components: a satellite selector and a task selector, working in conjunction to generate complete scheduling decisions.

The satellite selector leverages multi-head attention mechanisms to process both task-related and station-related features for each satellite. This component evaluates all available satellites based on their current states, encoded window features, and relational information to generate selection probabilities.

The task selector, activated after a satellite is chosen, processes the chosen satellite's state along with relevant observation target and station features to determine the most appropriate task for the selected satellite. It employs a unified selection mechanism that can handle both observation targets and ground stations as potential tasks, making it flexible for different operational scenarios.

This hierarchical structure allows the decoder to decompose the complex scheduling decision into manageable sub-decisions while maintaining the dependencies between satellite and task selection. In the following subsections, we detail the architecture and operation of each selector component.

5.4.1. Satellite selector

The satellite selector evaluates and selects satellites based on their current states and their relationships with potential targets and ground stations. It incorporates three main processing stages to generate satellite selection probabilities.

First, the selector generates unified feature representations for observation targets and ground stations through projection layers. For each satellite–target pair, it concatenates the target state, observation window features, and relational state, then projects them into a unified hidden dimension through:

$$\mathbf{h}_{ij}^{\text{target}} = W_{\text{proj}} \left[\mathbf{h}_j^T; \mathbf{h}_{ij}^{\text{obs}}; \mathbf{h}_{ij}^r \right] \quad (26)$$

where $\mathbf{h}_j^T$, $\mathbf{h}_{ij}^{\text{obs}}$, and $\mathbf{h}_{ij}^r$ represent the encoded target state, observation window features, and relational state respectively.

Similarly, for satellite–station pairs, station features are processed through:

$$\mathbf{h}_{ij}^{\text{station}} = W_{\text{proj}} \left[\mathbf{h}_j^G; \mathbf{h}_{ij}^{\text{trans}}; \mathbf{h}_{ij}^q \right] \quad (27)$$

where $\mathbf{h}_j^G$, $\mathbf{h}_{ij}^{\text{trans}}$, and $\mathbf{h}_{ij}^q$ denote the encoded station state, transmission window features, and relational state.

Second, the selector employs two parallel multi-head attention modules to generate target and station contexts for each satellite. Taking the satellite state as query and the projected features as key-value pairs:

$$\mathbf{c}_i^{\text{target}} = \text{MultiHeadAttn} \left(\mathbf{h}_i^S, \mathbf{h}_{ij}^{\text{target}}, \mathbf{h}_{ij}^{\text{target}} \right) \quad (28)$$

$$\mathbf{c}_i^{\text{station}} = \text{MultiHeadAttn} \left(\mathbf{h}_i^S, \mathbf{h}_{ij}^{\text{station}}, \mathbf{h}_{ij}^{\text{station}} \right) \quad (29)$$

where $\mathbf{c}_i^{\text{target}}$ and $\mathbf{c}_i^{\text{station}}$ capture the relevant target and station contexts for each satellite's decision-making.

Finally, the selector concatenates the satellite state with both target and station context and processes them through a fusion network:

$$\mathbf{h}_i^{\text{fused}} = \text{FFN}([\mathbf{h}_i^S; \mathbf{c}_i^{\text{target}}; \mathbf{c}_i^{\text{station}}]) \quad (30)$$

The fused features are then passed through a scoring network to generate selection probabilities:

$$p(i) = \text{softmax}(W_{\text{score}} \mathbf{h}_i^{\text{fused}}) \quad (31)$$

where $p(i)$ represents the probability of selecting satellite i . During inference, the satellite with the highest probability is selected, subject to the feasibility mask which ensures that only valid satellites can be chosen.

5.4.2. Task selector

The task selector determines the most appropriate task (either an observation target or a ground station) for the selected satellite. It employs a unified selection mechanism that processes both observation targets and station candidates through the same architecture, enabling flexible decision-making across different operation types.

Given the selected satellite's state $\mathbf{h}_i^S$ and the combined task features $\mathbf{H}^{\text{task}} = [\mathbf{h}^T; \mathbf{h}^G]$, the selector first applies multi-head attention to capture the relevance between the satellite and potential tasks:

$$\mathbf{c}_i^{\text{task}} = \text{MultiHeadAttn} \left(\mathbf{h}_i^S, \mathbf{H}^{\text{task}}, \mathbf{H}^{\text{task}} \right) \quad (32)$$

where $\mathbf{c}_i^{\text{task}}$ represents the context vector that encodes the satellite-specific task information for satellite i .

The context vector is then expanded and concatenated with the original task features to form combined representations:

$$\mathbf{h}_{j|i}^{\text{combined}} = [\mathbf{H}_j^{\text{task}}; \mathbf{c}_i^{\text{task}}] \quad (33)$$

The combined features are processed through a fusion network to capture the interactions between target features and contextual information:

$$\mathbf{h}_{j|i}^{\text{fused}} = \text{FFN}(\mathbf{h}_{j|i}^{\text{combined}}) \quad (34)$$

Finally, the selector generates selection probabilities through a scoring network:

$$p(j|i) = \text{softmax}(W_{\text{score}} \mathbf{h}_{j|i}^{\text{fused}}) \quad (35)$$

where $p(j|i)$ represents the probability of selecting task j given that satellite i has been chosen. The selection is subject to a feasibility mask that ensures only valid tasks (considering resource constraints and operational requirements) can be chosen.

5.4.3. Feasibility mask mechanism

To ensure valid scheduling decisions, UHTS-DRL employs a two-level feasibility mask mechanism that dynamically filters infeasible options at each decision step.

Satellite Selection Mask: At the first level, satellites are evaluated based on whether their current time t_i exceeds the planning horizon H . The satellite selection mask is defined as:

$$m_i^{\text{sat}} = \begin{cases} 1, & \text{if } t_i < H \\ 0, & \text{otherwise} \end{cases} \quad (36)$$

This ensures that only satellites within the operational timeframe remain selectable.

Task Selection Mask: For satellite i selected at the first level, task-level masks are generated based on three operation types:

Observation tasks: A target j is selectable only if it simultaneously satisfies five conditions:

$$m_{ij}^{\text{obs}} = \mathbb{I}(o_j = 0) \wedge \mathbb{I}(v_{ij} > 0) \wedge \mathbb{I}(\mu_j \leq m_i) \wedge \mathbb{I}(\delta_j \cdot E_{\text{rate}} \leq e_i) \wedge \mathbb{I}(\eta_{ij} < H) \quad (37)$$

where $\mathbb{I}(\cdot)$ is the indicator function, o_j denotes observation completion status, v_{ij} represents the number of remaining observation windows, μ_j is storage requirement, m_i is available memory, e_i is remaining energy, and η_{ij} is the earliest executable time.

Ground stations: A station k is selectable when:

$$m_{ik}^{\text{trans}} = \mathbb{I}(n_i > 0) \wedge \mathbb{I}(\kappa_{ik} > 0) \quad (38)$$

where n_i indicates the number of stored observed tasks and κ_{ik} denotes the number of remaining transmission windows.

Virtual task: This option is activated only when all other choices are infeasible:

$$m^{\text{virtual}} = \neg \left(\bigvee_j m_{ij}^{\text{obs}} \vee \bigvee_k m_{ik}^{\text{trans}} \right) \quad (39)$$

This ensures the scheduling continuity by allowing satellites to advance to the next period.

Dynamic Mask Updates: Masks are recalculated at each decision step based on updated states. After action execution, the transition function updates satellite resources (e_i , m_i , t_i), task completion flags (o_j , c_j), and relational states (η_{ij} , v_{ij} , ζ_{ik} , κ_{ik}). The earliest executable time η_{ij} is particularly crucial for handling overlapping time windows. For each satellite-task pair, η_{ij} is computed as:

$$\eta_{ij} = \min_{l \in W_{ij}} \left\{ \max(t_i, t_{\text{start}}^{ijl}) + \tau(\theta_i, \theta_{ijl}(t)) \right\} \quad (40)$$

where W_{ij} denotes available windows, t_{start}^{ijl} is the l -th window's start time, and $\tau(\theta_i, \theta_{ijl}(t))$ represents the attitude maneuvering time from current angles θ_i to required angles $\theta_{ijl}(t)$ at execution time t . For agile satellites, $\theta_{ijl}(t)$ is time-dependent as the required observation attitude varies with orbital position.

This formulation naturally addresses overlapping time windows by considering both temporal availability and time-dependent attitude constraints. When multiple tasks have overlapping time windows, their distinct maneuvering requirements at different orbital positions result in different earliest executable times, enabling accurate feasibility identification.

5.5. Training algorithm

To train our policy network efficiently, we propose [Algorithm 1](#) which employs the REINFORCE algorithm with baseline. The algorithm integrates multiple mechanisms including baseline reward estimation, entropy regularization, and statistical significance testing.

At each iteration, we first generate a batch of problem instances and construct solutions for them using the current policy network π_θ . For each solution construction, the baseline policy network computes a baseline reward using a deterministic greedy strategy. The policy parameters are then updated using:

$$\nabla_\theta J(\theta) = \frac{1}{B} \sum_{b=1}^B (R_b - R_b^{\text{BL}}) \nabla_\theta \log \pi_\theta(a_t | s_t) \quad (41)$$

where B is the batch size, R_b is the actual reward obtained by policy π_θ , and R_b^{BL} is the baseline reward. The parameters are optimized using the Adam optimizer. To reduce variance further, the baseline policy is updated only when the current policy shows statistically significant improvement according to a one-sided paired t -test with significance level α .

6. Experimental results and discussions

In this section, we conduct extensive simulation experiments designed to validate the efficiency of our proposed method through comparisons with both heuristic and meta-heuristic approaches. [Section 6.1](#) details the scenario settings, while [Section 6.2](#) outlines the algorithm configurations. Comparative results for various task scales are presented in [Section 6.3](#); the impact of different resource configurations is analyzed in [Section 6.4](#); and the performance across various target distributions is illustrated in [Section 6.5](#).

Algorithm 1: Training method.

Input: Policy network π_θ parameters θ ; baseline network v_ϕ parameters ϕ ; number of epochs E ; epoch size N ; number of batches B , significance level α

Output: Trained policy network

```

1 for  $epoch = 1, 2, \dots, E$  do
2   Sample  $N$  problem instances randomly;
3   for each batch  $B$  do
4      $done = \text{False}$ ;
5     while  $\neg done$  do
6       Pick an action  $(i, j) \sim \pi_\theta(a_t | s_t)$ ;
7       Observe reward  $r_t$  and next state  $s_t$ ;
8       if reach the terminate condition then
9          $done = \text{True}$ ;
10   $R_b = \sum_t r_t$ ;
11  GreedyRollout with baseline and obtain its reward  $R_b^{BL}$ ;
12   $\nabla_\theta J(\theta) = \frac{1}{B} \sum_{b=1}^B (R_b - R_b^{BL}) \nabla_\theta \log \pi_\theta(a_t | s_t)$ ;
13   $\theta \leftarrow \text{Adam}(\theta, \nabla_\theta J(\theta))$ ;
14  if  $\text{OneSidedPairedTTest}(\pi_\theta, v_\phi) < \alpha$  then
15     $\phi \leftarrow \theta$ ;

```

To evaluate the performance of different algorithms, we employ the following metrics:

- **Completed Tasks:** The number of tasks with both observation and transmission completed.
- **Task Completion Rate:** The percentage of completed tasks, calculated as (Number of Completed Tasks / Total Number of Tasks) $\times 100\%$.
- **Total Profit:** The sum of profits from all completed and partially completed tasks, where fully completed tasks receive full profit π_j , while observed-only tasks receive partial profit $w_j^o \cdot \pi_j$.
- **Profit Rate:** The percentage of maximum achievable profit obtained, calculated as (Obtained Total Profit / Maximum Total Profit) $\times 100\%$, where Maximum Total Profit represents the sum of all task profits in the scenario.
- **Relative Profit Improvement:** The performance improvement of UHTS-DRL over baseline methods in terms of total profit, calculated as (Profit_UHTS-DRL - Profit_baseline) / Profit_baseline $\times 100\%$.

6.1. Simulation scenario setting

The simulation scenarios are designed with a constellation consisting of six sun synchronous orbit satellites. The orbit parameters are shown in Table 6. Five well-known ground stations within Chinese territory were selected, which are located in the south, west, north, east and center of China. Each ground station is equipped with one antenna for receiving the data from satellites. In addition, the scheduling period was set from 00:00:00 on September 22, 2024, to 12:00:00 on September 22, 2024, covering a planning horizon of 12 h [30].

To construct the training dataset, we randomly sampled 2000 target points within the latitude range of $[-80^\circ, 80^\circ]$ and longitude range of $[-168^\circ, 168^\circ]$. For each target, the profit value was randomly assigned between 1 and 10, and the observation duration was randomly set between 5 and 10 s. The observation time windows and transmission time windows can be pre-calculated using either fast visibility algorithms [38,39] or the software Satellite Tool Kit (STK). During the training process, a batch of targets was

Table 6
Orbit parameters for each satellite.

Satellite parameters	Satellite 1	Satellite 2	Satellite 3	Satellite 4	Satellite 5	Satellite 6
Orbit semi-major axis(km)	7163.14	7163.14	7163.14	7163.14	7163.14	7163.14
Eccentricity($^\circ$)	0.00013	0.00013	0.00013	0.00013	0.00013	0.00013
Inclination($^\circ$)	98.5	98.5	98.5	98.5	98.5	98.5
True anomaly($^\circ$)	10	20	30	40	50	60
Argument of perigee($^\circ$)	0	0	0	0	0	0
RAAN($^\circ$)	172.85	142.85	112.85	82.85	52.85	22.85
Maximum maneuvering angle($^\circ$)	45	45	45	45	45	45
Angular velocity($^\circ$ /s)	2.0	2.0	2.0	2.0	2.0	2.0

randomly selected from this dataset for each training iteration. This sampling approach ensures sufficient diversity in the training data to enhance the model's generalization capability.

First, we trained our UHTS-DRL model under three basic scenarios with 6 satellites and 5 ground stations, where the number of targets was set to 20, 50, and 100 respectively. These three models served as our base models to evaluate the convergence capability of our approach. After training, the model trained with 100 targets was used for testing across all experimental scenarios.

To comprehensively evaluate the performance and generalization capability of our proposed approach, we designed three series of experiments. First, we conducted scalability experiments by testing different task scales ranging from 20 to 400 targets while maintaining 6 satellites and 5 ground stations, aiming to verify the algorithm's ability to handle increasing problem sizes. Second, we examined the impact of resource configurations by varying the number of satellites (4/5/6/7/8) and ground stations (3/4/5) while fixing the number of tasks at 200; although our algorithm can scale to handle an arbitrary number of ground stations, we strategically employed five real-world operational stations and tested with reduced configurations to validate performance under limited resource scenarios. Finally, we investigated the algorithm's adaptability to different spatial distributions of targets. The detailed experimental configurations are shown in Tables 7–9.

6.2. Algorithm setting

The hyperparameters of UHTS-DRL are summarized in Table 10. We used different batch sizes for different problem scales: 256 for 20/50-target scenarios and 128 for 100-target scenarios. This adjustment is necessary due to GPU memory constraints, as larger problem scales require more memory per instance for state representations and neural network computations. Reducing the batch size for 100-target problems ensures stable training on the NVIDIA A100 GPU while maintaining sufficient samples for effective gradient estimation. All other hyperparameters remained consistent across training scenarios.

Table 7
Task scale scalability experiment configurations.

Scenario	Target number	Satellite number	Ground station number
SCEN_1	20	6	5
SCEN_2	50	6	5
SCEN_3	100	6	5
SCEN_4	150	6	5
SCEN_5	200	6	5
SCEN_6	250	6	5
SCEN_7	300	6	5
SCEN_8	350	6	5
SCEN_9	400	6	5

Table 8
Resource scalability experiment configurations.

Scenario	Target number	Satellite number	Ground station number
SCEN_10	200	4	3
SCEN_11	200	4	4
SCEN_12	200	4	5
SCEN_13	200	5	3
SCEN_14	200	5	4
SCEN_15	200	5	5
SCEN_16	200	6	3
SCEN_17	200	6	4
SCEN_18	200	6	5
SCEN_19	200	7	3
SCEN_20	200	7	4
SCEN_21	200	7	5
SCEN_22	200	8	3
SCEN_23	200	8	4
SCEN_24	200	8	5

Table 9
Spatial distribution experiment configurations.

Scenario	Distribution pattern	Target region	Description
SCEN_25	Uniform	Global	Targets uniformly distributed across $[-80^\circ, 80^\circ] \times [-168^\circ, 168^\circ]$
SCEN_26	Clustered	Three Clusters	Three high-density regions with specified cluster centers
SCEN_27	Latitude-biased	Low Latitude	Higher target density between $[-30^\circ, 30^\circ]$ latitude
SCEN_28	Longitude-biased	Eastern Hemisphere	Higher target density in $[0^\circ, 168^\circ]$ longitude
SCEN_29	Mixed	Global	Combination of uniform background with two high-density clusters

Table 10
Hyperparameter settings for UHTS-DRL.

Parameter	Value
Attention heads	8
Hidden dimension	64
Number of epochs	200
Batch size (20/50 targets)	256
Batch size (100 targets)	128
Learning rate	0.0001
Max gradient norm	1
Significance level	0.05
Lower weight profit for observation task w_j^o	0.2

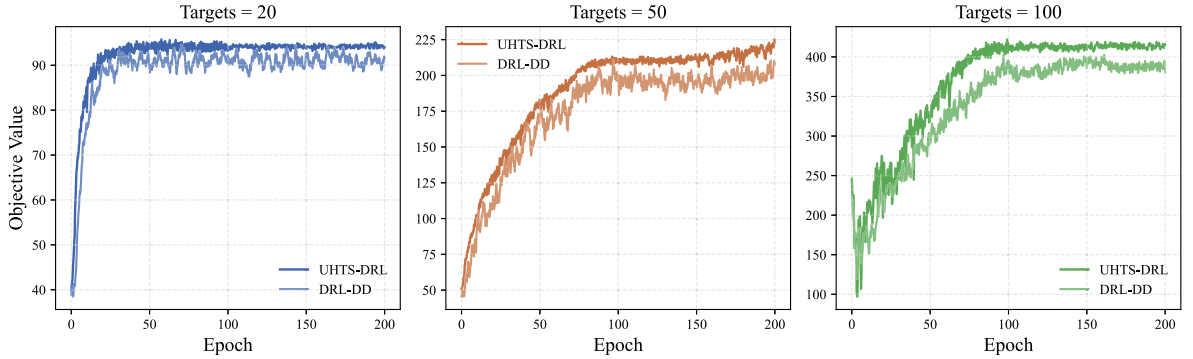


Fig. 4. The training convergence curve.

We compared UHTS-DRL with several representative baseline algorithms: (1) Priority-based heuristic algorithm (PBHA) [40], which schedules tasks based on their priority values; (2) Genetic algorithm (GA) [7] which encodes both task sequence and resource allocation; (3) A two-phase genetic annealing algorithm (TPGA) [41]—all three originally designed for non-agile satellite scheduling but modified to handle agile satellites by setting task start times to their earliest feasible times; (4) Large neighborhood search algorithm (LNS) [42], which employs GA's encoding scheme but uses random destruction and repair operators; (5) PBHA_adjust and (6) GA_adjust [43], which incorporate task execution time adjustment mechanisms; and (7) A DRL method based on dual-decoder architecture (DRL-DD) [30], originally proposed for data relay satellite network task scheduling, which we adapted to our unified MDP formulation to handle the MSIOTSP.

Before presenting the detailed experimental results, we first analyze the convergence behavior of UHTS-DRL compared with DRL-DD. Fig. 4 shows the training curves under three different problem scales (20, 50, and 100 targets). Both methods exhibit clear convergence patterns with rapid improvement in early training followed by stabilization. However, UHTS-DRL consistently outperforms DRL-DD across all problem scales, with the performance gap becoming more pronounced as problem size increases. Notably, UHTS-DRL maintains higher objective values throughout the training process and demonstrates more stable convergence behavior with less fluctuation in later epochs. This superior performance demonstrates that our specialized LSTM-based time window encoder and hierarchical dual-decoder architecture, designed specifically for integrated observation-transmission scheduling, are more effective than the general-purpose dual-decoder framework adapted from relay satellite networks.

6.3. Task size difference analysis

We first evaluate our proposed method against baseline approaches across different problem scales. For comprehensive comparison, we conduct extensive experiments on 7 scenarios (SCEN_1 to SCEN_7) with target numbers ranging from 20 to 400, while maintaining 6 satellites and 5 ground stations.

Table 11 presents the detailed comparison results in terms of the number of completed tasks, task completion rate, total obtained profit, profit rate and computation time. For small-scale problems (20-50 tasks), all algorithms demonstrate comparable performance with minimal differences in total profit. However, as the problem scale increases, significant performance disparities emerge. UHTS-DRL consistently outperforms all baseline algorithms across most medium and large-scale scenarios.

Traditional algorithms demonstrate relatively stable performance but fail to capitalize effectively on the growing solution space as problem complexity increases. In contrast, UHTS-DRL significantly outperforms the comparison algorithms in terms of solution quality. Specifically, in SCEN_6, UHTS-DRL achieves a total profit of 810.8, exhibiting a 11.7% relative profit improvement over TPGA and 12.5% over GA_adjust. Compared to the learning-based baseline DRL-DD, UHTS-DRL also demonstrates superior performance across all problem scales, with the advantage becoming more pronounced as problem size increases. Notably, the UHTS-DRL method obtained superior solutions with remarkable computational efficiency, achieving a better balance between solution quality

Table 11
Experimental results of different task sizes.

Method	Metric	SCEN_1	SCEN_2	SCEN_3	SCEN_4	SCEN_5	SCEN_6	SCEN_7	SCEN_8	SCEN_9
PBHA	Completed Tasks	19	43	75	71	72	73	77	77	77
	Completion Rate (%)	95.00	86.00	75.00	47.33	36.00	29.20	25.67	22.00	19.25
	Total Profit	81.00	222.00	425.00	548.60	608.20	630.40	692.40	710.80	719.20
	Profit Rate (%)	90.00	88.80	83.50	68.59	56.56	47.55	43.55	38.01	33.84
	Computation Time (s)	0.02	0.06	0.13	0.27	0.35	0.63	0.53	0.61	0.71
GA	Completed Tasks	19	43	87	114	118	125	124	125	135
	Completion Rate (%)	95.00	86.00	87.00	76.00	59.00	50.00	41.33	35.71	33.75
	Total Profit	81.00	223.00	442.20	630.20	655.60	663.40	665.40	727.60	738.80
	Profit Rate (%)	90.00	89.20	86.84	78.85	60.93	50.04	41.85	38.92	34.73
TPGA	Computation Time (s)	10.44	24.99	59.96	95.62	127.69	142.66	177.55	200.34	304.81
	Completed Tasks	19	43	86	89	87	93	83	87	83
	Completion Rate (%)	95.00	86.00	86.00	59.33	43.50	37.20	27.67	24.86	20.75
	Total Profit	81.00	222.00	444.40	617.40	678.20	725.40	736.80	782.80	765.00
	Profit Rate (%)	90.00	88.80	87.23	77.22	63.07	54.72	46.32	41.86	36.00
LNS	Computation Time (s)	48.66	169.66	217.29	283.25	343.41	263.26	252.28	288.91	282.35
	Completed Tasks	19	43	77	73	74	75	79	79	79
	Completion Rate (%)	95.00	86.00	77.00	48.67	37.00	30.00	26.33	22.57	19.75
	Total Profit	81.00	222.00	429.40	559.80	621.20	646.40	707.60	726.80	737.00
	Profit Rate (%)	90.00	88.80	84.28	69.96	57.77	48.75	44.49	38.87	34.64
PBHA_adjust	Computation Time (s)	0.10	0.24	0.54	2.11	2.36	0.73	2.70	3.73	2.37
	Completed Tasks	19	43	74	72	74	75	74	76	76
	Completion Rate (%)	95.00	86.00	74.00	48.00	37.00	30.00	24.67	21.71	19.00
	Total Profit	81.00	222.00	424.60	555.20	622.20	649.60	669.80	702.20	715.00
	Profit Rate (%)	90.00	88.80	83.30	69.46	57.86	48.98	42.10	37.58	33.65
GA_adjust	Computation Time (s)	0.02	0.06	0.14	0.23	0.33	0.40	0.47	0.60	0.70
	Completed Tasks	19	43	86	111	125	128	129	125	130
	Completion Rate (%)	95.00	86.00	86.00	74.00	62.50	51.20	43.00	35.71	32.50
	Total Profit	82.80	222.00	449.40	592.20	666.20	720.60	721.60	676.40	741.20
	Profit Rate (%)	92.00	88.80	88.21	74.09	61.95	54.34	45.37	36.19	34.87
DRL-DD	Computation Time (s)	11.18	28.37	58.53	154.66	113.73	138.47	196.28	300.69	327.20
	Completed Tasks	19	44	87	108	116	119	120	126	127
	Completion Rate (%)	95.00	88.00	87.00	72.00	58.00	47.60	40.00	36.00	31.75
	Total Profit	82.80	223.20	446.60	585.40	658.40	692.20	723.60	695.60	743.40
	Profit Rate (%)	92.00	89.28	87.66	73.24	61.22	52.20	45.50	37.22	34.97
UHTS-DRL	Computation Time (s)	0.79	0.99	1.54	2.36	2.43	2.87	3.02	3.13	3.20
	Completed Tasks	19	44	88	118	118	147	151	150	134
	Completion Rate (%)	95.00	88.00	88.00	78.67	59.00	58.80	50.33	42.86	33.50
	Total Profit	82.80	229.80	459.80	661.00	687.80	810.80	839.60	841.40	785.20
	Profit Rate (%)	92.00	91.92	90.33	82.73	63.98	61.19	52.84	45.04	36.95
	Computation Time (s)	0.89	1.13	1.77	2.42	2.38	3.08	3.14	3.29	3.15

and computational efficiency. The superior scalability of UHTS-DRL can be attributed to its ability to learn effective scheduling patterns through reinforcement learning, which enables better generalization across problem scales and more efficient exploitation of the expanded solution space in large-scale scenarios. From Fig. 5, the UHTS-DRL can consistently achieve higher total profit and complete more tasks across different problem scales, and the advantage of the UHTS-DRL is also evident under large-scale problems with 350 and 400 tasks.

6.4. Resource scale difference analysis

The experimental results across different resource configurations are shown in Fig. 6. The results clearly demonstrate the superior performance of the UHTS-DRL algorithm. UHTS-DRL consistently outperforms the baseline algorithms across most scenarios. For example, when only 3 ground stations are available, UHTS-DRL achieves up to 53.28% profit rate with 8 satellites compared to 49.12%-52.19% for other methods. Compared to DRL-DD, UHTS-DRL also shows consistent improvements across all resource configurations. The performance gap between UHTS-DRL and baseline algorithms widens further as ground station resources increase. Specifically, UHTS-DRL reaches 86.36% profit rate in the maximum resource configuration which contains 8 satellites and 5 ground stations, demonstrating its adaptability to resource scaling.

Fig. 6 also illustrates the trend of profit rate under different resource configurations. The profit rate shows a significant upward trend as resources increase, with all algorithms benefiting from additional satellites or ground stations. For satellite scaling, profit rates increase by approximately 20%-30% when moving from 4 to 8 satellites. Similarly, ground station scaling shows more improvements, with profit rates increasing by 25%-35% when expanding from 3 to 5 ground stations for any fixed number of satellites. This pattern

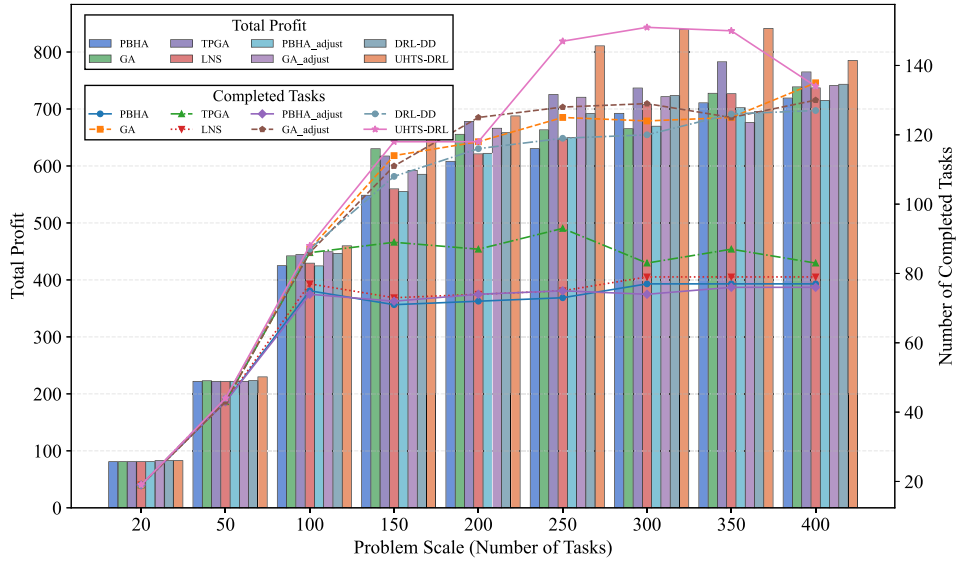


Fig. 5. Comparison of profits and number of tasks completed under different sizes of tasks.

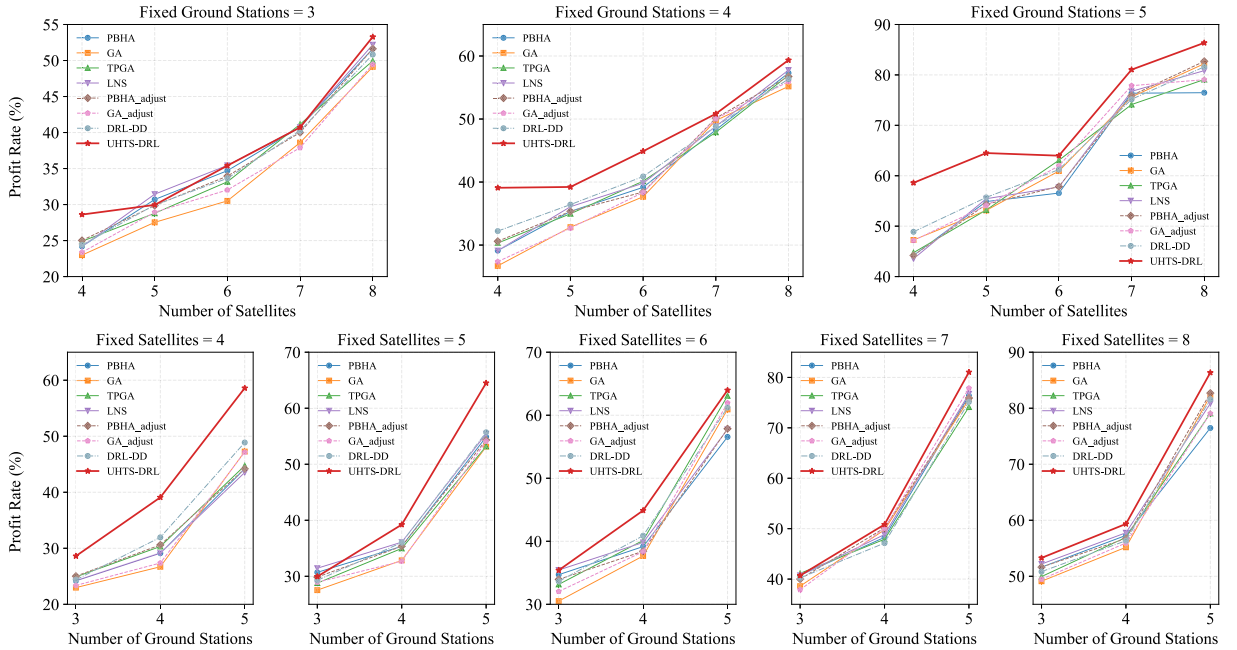


Fig. 6. Profit rate trends across various resource configurations.

highlights the critical importance of ground station availability for maximizing system performance. To rigorously validate the relative impact of satellite and ground station resources on system performance, we conducted a two-way analysis of variance (ANOVA) [44] using the 15 resource configurations from SCEN_10 to SCEN_24 with profit rate as the dependent variable. Both factors exhibit highly significant effects at the $p < 0.001$ level. Ground stations demonstrate a substantially stronger impact, with an F -statistic of 187.34 compared to 38.32 for satellites, confirming that ground station availability represents a more critical bottleneck in the integrated scheduling problem (Table 12).

6.5. Target distribution difference analysis

To evaluate the adaptability of the proposed UHTS-DRL across varying spatial distributions of targets, we conduct a series of experiments under five distinct target distributions. The target distributions are shown in Fig. 7. Specifically, in SCEN_25, targets are sampled uniformly across latitudes $[-80^\circ, 80^\circ]$ and longitudes $[-168^\circ, 168^\circ]$; in SCEN_26, they are organized into three regional clusters

Table 12
Two-way ANOVA results for resource configuration analysis.

Source	<i>F</i>	<i>p</i> -value	Partial η^2
Satellite Count	38.32	<0.001	0.95
Ground Station Count	187.34	<0.001	0.979

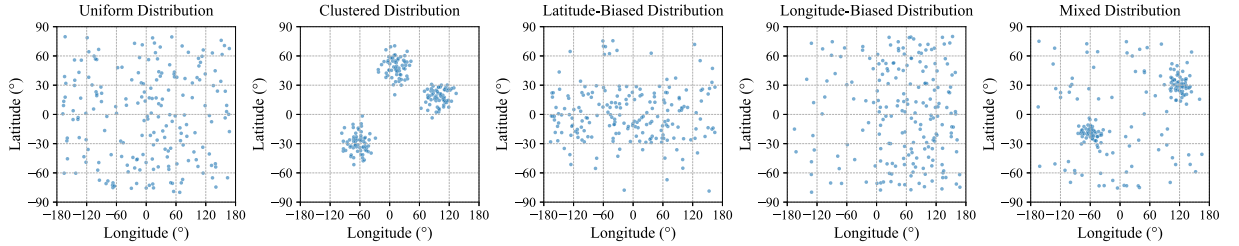


Fig. 7. Different spatial distribution patterns of targets.

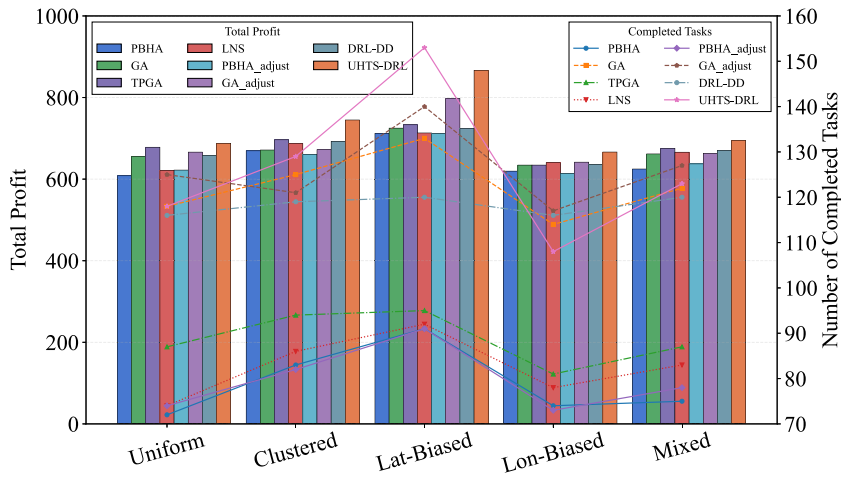


Fig. 8. Comparison of profits and number of tasks completed under different target distributions.

drawn from normal distributions with standard deviations of 10° in latitude and 15° in longitude. SCEN_27 introduces a latitude-biased distribution in which 70% of targets are placed within the equatorial band $[-30^\circ, 30^\circ]$, while the remainder are distributed globally. In SCEN_28, 70% of targets are concentrated in the Eastern Hemisphere $[0^\circ, 168^\circ]$. Finally, SCEN_29 employs a mixed configuration consisting of a uniformly distributed background accounting for 60% of targets and two high-density clusters, each containing 20% of targets sampled from normal distributions with latitude and longitude standard deviations of 8° and 12° , respectively.

The experimental results are shown in Fig. 8. The results demonstrate that UHTS-DRL consistently achieves superior performance across various geographical distributions, with total profits ranging from 666.40 to 866.80 and completed tasks ranging from 108 to 153. In contrast, baseline methods exhibit lower performance, with PBHA achieving profits from 608.20 to 712.20 and completed tasks from 72 to 91, GA achieving profits from 634.40 to 725.00 and completed tasks from 114 to 133, and TPGA achieving profits from 634.40 to 733.20 and completed tasks from 81 to 95. Compared to DRL-DD, UHTS-DRL also demonstrates notable advantages across different target distributions, achieving higher total profits in all scenarios such as 687.80 versus 658.40 in uniform distribution and 866.80 versus 723.60 in latitude-biased distribution, while maintaining comparable or higher task completion numbers. This consistent superiority across various geographical distributions proves the algorithm's strong adaptability and reliability in satellite task scheduling problems.

7. Conclusion

In this paper, we addressed the Multi-agile Satellite and Multi-ground Station Integrated Observation and Data Transmission Scheduling Problem (MSIODTSP). We proposed a unified Markov Decision Process model that successfully integrates both observation and transmission scheduling decisions. Based on the proposed MDP, we developed UHTS-DRL, an end-to-end framework that utilizes a hierarchical neural network architecture with three key components: a time window encoder that processes time window constraints, a state encoder capturing dynamic system states, and a two-stage scheduling decoder employing attention mechanisms for satellite and task selection. Our comprehensive experiments demonstrated that UHTS-DRL consistently outperforms traditional scheduling

methods across various scenarios. The approach achieved superior performance, particularly in large-scale problems, with up to 12.5% relative improvement in profit rates. The results confirmed UHTS-DRL's adaptability to different problem scales, resource configurations, and geographical target distributions while maintaining computational efficiency.

While our approach demonstrates strong performance, there remain opportunities for further refinement in several aspects. The current framework adopts simplified modeling assumptions, including linear energy consumption dynamics, abstracted communication link characteristics, and a single-objective optimization formulation. For future research, we will extend UHTS-DRL to handle more complex real-world constraints such as weather uncertainty, onboard processing capabilities and other dynamic disturbance factors [37,45]. For extremely large-scale scenarios, we will explore multi-scale training strategies and curriculum learning to further enhance generalization capability. We will also investigate multi-agent reinforcement learning approaches to enable decentralized decision-making across satellite constellations, incorporate multi-objective optimization frameworks to balance competing objectives such as profit maximization, energy efficiency, and mission fairness, and explore the application of our approach to heterogeneous satellite constellations with varying capabilities and orbit types.

CRedit authorship contribution statement

Jiaqi Cheng: Writing – original draft, Software, Resources, Methodology, Investigation, Data curation, Conceptualization. **Mingfeng Fan:** Writing – review & editing, Software, Methodology. **Yi Gu:** Writing – review & editing, Data curation. **Wei Tang:** Resources, Data curation. **Qizhang Luo:** Writing – review & editing, Data curation. **Yalin Wang:** Writing – review & editing, Resources. **Xinwei Wang:** Writing – review & editing, Supervision. **Guohua Wu:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The author is an Editorial Board Member/Editor-in-Chief/Associate Editor/Guest Editor for this journal and was not involved in the editorial review or the decision to publish this article.

Data availability

Data will be made available on request.

References

- [1] P. Wang, H. Li, S. Zhang, X. Lin, J. Liu, E. Wang, A novel joint scheduling scheme of earth observation and transmission in satellite networks, in: 2020 International Conference on Computing, Networking and Communications (ICNC), IEEE, 2020, pp. 774–779.
- [2] L. He, B. Liang, J. Li, M. Sheng, Joint observation and transmission scheduling in agile satellite networks, *IEEE Trans. Mob. Comput.* 21 (2021) 4381–4396.
- [3] X. Wang, G. Wu, L. Xing, W. Pedrycz, Agile earth observation satellite scheduling over 20 years: formulations, methods, and future directions, *IEEE Syst. J.* 15 (2020) 3881–3892.
- [4] M. Chen, B. Li, L. Xing, Y. Chen, DEEM: a differential evolution ensemble method for optimizing agile satellite scheduling, *Expert Syst. Appl.* 272 (2025) 126771.
- [5] L. Wei, Y. Cui, M. Chen, Q. Wan, L. Xing, Multi-objective neural policy approach for agile earth satellite scheduling problem considering image quality, *Swarm Evol. Comput.* 94 (2025) 101857.
- [6] Y. Ren, Q. Li, Y. Guo, W. Pedrycz, L. Xing, A. Liu, Y. Song, A distance similarity-based genetic optimization algorithm for satellite ground network planning considering feeding mode, *Expert Syst. Appl.* 268 (2025) 126303.
- [7] J. Zhang, L. Xing, An improved genetic algorithm for the integrated satellite imaging and data transmission scheduling problem, *Comput. Oper. Res.* 139 (2022) 105626.
- [8] X. Hu, W. Zhu, B. An, P. Jin, W. Xia, A branch and price algorithm for EOS constellation imaging and downloading integrated scheduling problem, *Comput. Oper. Res.* 104 (2019) 74–89.
- [9] S. Peng, H. Chen, J. Li, N. Jing, A heuristic method for single satellite observation and transmission tasks planning, in: 2017 13th International Conference on Natural Computation, Fuzzy Systems and Knowledge Discovery (Icnc-Fskd), IEEE, 2017, pp. 2925–2932.
- [10] Y. Wang, M. Sheng, W. Zhuang, S. Zhang, N. Zhang, J. Li, Joint scheduling of observation and transmission in earth observation satellite networks, in: GLOBECOM 2017-2017 IEEE Global Communications Conference, IEEE, 2017, pp. 1–6.
- [11] M. Shang, R. Yuan, B. Song, X. Huang, B. Yang, S. Li, Joint observation and transmission scheduling of multiple agile satellites with energy constraint using improved ACO algorithm, *Acta Astronaut.* 230 (2025) 92–103.
- [12] J. Cheng, M. Fan, X. Zhang, J. Liang, Y. Cao, G. Wu, G.A. Sartoretti, Multimodal fused learning for solving the generalized traveling salesman problem in robotic task planning, *arXiv preprint arXiv:2506.16931*, 2025.
- [13] M. Fan, Y. Wu, Z. Cao, W. Song, G. Sartoretti, H. Liu, G. Wu, Conditional neural heuristic for multiobjective vehicle routing problems, *IEEE Trans. Neural Netw. Learn. Syst.* 36 (2024) 4677–4689.
- [14] X. Wang, F. Zhao, Z. Shi, Z. Jin, Deep reinforcement learning-based periodic earth observation scheduling for agile satellite constellation, *J. Aerosp. Inf. Syst.* 20 (2023) 508–519.
- [15] K. Chen, S. Du, H. Huo, T. Fang, An integrated scheduling algorithm of satellite observation and data download based on multi-agent deep reinforcement learning, in: 2023 China Automation Congress (CAC), IEEE, 2023, pp. 3234–3239.
- [16] P. Li, H. Wang, Y. Zhang, R. Pan, Mission planning for distributed multiple agile earth observing satellites by attention-based deep reinforcement learning method, *Adv. Space Res.* 74 (5) (2024) 2388–2404.
- [17] Y. He, L. Xing, Y. Chen, W. Pedrycz, L. Wang, G. Wu, A generic markov decision process model and reinforcement learning method for scheduling agile earth observation satellites, *IEEE Trans. Syst. Man Cybern. Syst.* 52 (2020) 1463–1474.
- [18] Z. Liu, W. Xiong, C. Han, X. Yu, Deep reinforcement learning with local attention for single agile optical satellite scheduling problem, *Sens.* 24 (2024) 6396.
- [19] C. Chen, N. Wang, R. He, Research on earth observation satellite scheduling based on deep reinforcement learning and rule method, in: 2024 36th Chinese Control and Decision Conference (CCDC), IEEE, 2024, pp. 2397–2401.
- [20] S.H.I. Qi, L.I. Lu, F.A.N.G. Ziruo, B.I. Xingzi, L.L.U. Huaqiu, X. Zhang, C.H.E.N. Wen, Y.U. Jinpei, Efficient and fair PPO-based integrated scheduling method for multiple tasks of SATech-01 satellite, *Chin. J. Aeronaut.* 37 (2024) 417–430.
- [21] Z. Wang, X. Hu, H. Ma, W. Xia, Learning multi-satellite scheduling policy with heterogeneous graph neural network, *Adv. Space Res.* 73 (2024) 2921–2938.
- [22] Z. Liu, W. Xiong, C. Han, K. Zhao, A double-pointer network for multiple agile optical satellite scheduling problem, *Aerospace* 11 (2024) 792.
- [23] A. Herrmann, M.A. Stephenson, H. Schaub, Single-agent reinforcement learning for scalable earth-observing satellite constellation operations, *J. Spacecr. Rockets* 61 (2024) 114–132.

- [24] Y. Song, J. Ou, W. Pedrycz, P.N. Suganthan, X. Wang, L. Xing, Y. Zhang, Generalized model and deep reinforcement learning-based evolutionary method for multitype satellite observation scheduling, *IEEE Trans. Syst. Man Cybern. Syst.* 54 (4) (2024) 2576–2589.
- [25] Y. Chen, X. Shen, G. Zhang, Z. Lu, Multi-objective multi-satellite imaging mission planning algorithm for regional mapping based on deep reinforcement learning, *Remote Sens.* 15 (2023) 3932.
- [26] Z. Wen, Y. Liu, S. Zhang, H. Hu, Scheduling observation tasks for large-scale satellite constellation, in: *Journal of Physics: Conference Series*, vol. 2746, IOP Publishing, 2024, p. 012040.
- [27] X. Wang, J. Wu, Z. Shi, F. Zhao, Z. Jin, Deep reinforcement learning-based autonomous mission planning method for high and low orbit multiple agile earth observing satellites, *Adv. Space Res.* 70 (2022) 3478–3493.
- [28] B. Ren, Z. Zhu, F. Yang, T. Wu, H. Yuan, High-altitude satellites range scheduling for urgent request utilizing reinforcement learning, *Open Astron.* 31 (2022) 268–275.
- [29] J. Ou, L. Xing, F. Yao, M. Li, J. Lv, Y. He, Y. Song, J. Wu, G. Zhang, Deep reinforcement learning method for satellite range scheduling problem, *Swarm Evol. Comput.* 77 (2023) 101233.
- [30] J. Li, G. Wu, T. Liao, M. Fan, X. Mao, W. Pedrycz, Task scheduling under a novel framework for data relay satellite network via deep reinforcement learning, *IEEE Trans. Veh. Technol.* 72 (2023) 6654–6668.
- [31] C. Ma, D. Zhou, M. Sheng, H. Li, J. Li, Dynamic TT&C mission scheduling for mega-satellite networks: a deep reinforcement learning approach, in: *GLOBECOM 2023-2023 IEEE Global Communications Conference*, IEEE, 2023, pp. 7163–7168.
- [32] K. Li, S. Xiong, M. Yang, S. Ma, Online assignment of satellite data transmission tasks based on twin delayed deep deterministic policy gradient, in: *2024 7th World Conference on Computing and Communication Technologies (WCCCT)*, IEEE, 2024, pp. 245–249.
- [33] C. Weihong, X. Bowen, S. Jicheng, W. Zheng, G. Congcong, L. Shichao, Reinforcement learning-based ground stations scheduling algorithm for leo satellites, in: *Seventh International Conference on Electromechanical Control Technology and Transportation (ICECTT 2022)*, vol. 12302, SPIE, 2022, pp. 1034–1046.
- [34] Q. Qu, K. Liu, X. Li, Y. Zhou, J. Lü, Satellite observation and data-transmission scheduling using imitation learning based on mixed integer linear programming, *IEEE Trans. Aerosp. Electron. Syst.* 59 (2022) 1989–2001.
- [35] Z. Wen, L. Li, J. Song, S. Zhang, H. Hu, Scheduling single-satellite observation and transmission tasks by using hybrid actor-critic reinforcement learning, *Adv. Space Res.* 71 (2023) 3883–3896.
- [36] Y. Long, C. Shan, W. Shang, J. Li, Y. Wang, Deep reinforcement learning-based approach with varying-scale generalization for the earth observation satellite scheduling problem considering resource consumptions and supplements, *IEEE Trans. Aerosp. Electron. Syst.* 60 (3) (2024) 2572–2585.
- [37] X. Wang, G. Song, R. Leus, C. Han, Robust earth observation satellite scheduling with uncertainty of cloud coverage, *IEEE Trans. Aerosp. Electron. Syst.* 56 (2019) 2450–2461.
- [38] Y. Gu, C. Han, X. Wang, A kriging based framework for rapid satellite-to-site visibility determination, in: *2019 IEEE 10th International Conference on Mechanical and Aerospace Engineering (ICMAE)*, IEEE, 2019, pp. 262–267.
- [39] X. Wang, C. Han, P. Yang, X. Sun, Onboard satellite visibility prediction using metamodeling based framework, *Aerosp. Sci. Technol.* 94 (2019) 105377.
- [40] Y. Wang, M. Sheng, W. Zhuang, S. Zhang, N. Zhang, R. Liu, J. Li, Multi-resource coordinate scheduling for earth observation in space information networks, *IEEE J. Sel. Areas Commun.* 36 (2018) 268–279.
- [41] Z. Waiming, H. Xiaoxuan, X. Wei, J. Peng, A two-phase genetic annealing method for integrated earth observation satellite scheduling problems, *Soft Comput.* 23 (2019) 181–196.
- [42] L. He, M. de Weerd, N. Yorke-Smith, Time/sequence-dependent scheduling: the design and evaluation of a general purpose tabu-based adaptive large neighbourhood search algorithm, *J. Intell. Manuf.* 31 (2020) 1051–1078.
- [43] B. Deng, C. Jiang, J. Wang, L. Kuang, Beam scheduling with various mission demands in data relay satellite systems, *J. Commun. Inf. Netw.* 6 (2021) 396–410.
- [44] D.C. Montgomery, *Design and Analysis of Experiments*, John Wiley & Sons, 2017.
- [45] C. Han, Y. Gu, G. Wu, X. Wang, Simulated annealing-based heuristic for multiple agile satellites scheduling under cloud coverage uncertainty, *IEEE Trans. Syst. Man Cybern. Syst.* 53 (2022) 2863–2874.